GLM Analysis: Logistic Regression

Single observer (GS) · n = 2051 valid trials · 4 sessions

Asset: Exp_DecoupledDots_005m + Exp_DecoupledDots_Inv_005m

Sessions: S1 (260406_1532) · S2-inv (260406_1754) · S3-inv (260407_0643) · S4 (260407_0731)

Predictors

F1 **dot-onset cueing**

1 = delayed-onset field translates (cued); 0 = non-delayed field translates (uncued)

F2 **depth-field cueing**

1 = translating field's depth-plane identity matches the delayed-onset field (depth-cued); 0 = depth-uncued

F3 **color-field cueing**

1 = translating field's color identity matches the delayed-onset field (color-cued); 0 = color-uncued

F4 **translator depth plane**

1 = Far; 0 = Near [covariate; see rationale]

$$\text{log-odds(correct)} = \beta_0 + \beta_1\cdot F1 + \beta_2\cdot F2 + \beta_3\cdot F3 + \beta_4\cdot F4 + \beta_{12}\cdot(F1\times F2) + \beta_{14}\cdot(F1\times F4) + \beta_{24}\cdot(F2\times F4)$$

Rationale

Trial-level accuracy (correct/incorrect) was modeled using binary logistic regression. Three predictors were specified a priori by the experimental design. F1 (dot-onset cueing) coded whether the field with the delayed onset was the translating field (1 = cued, 0 = uncued). F2 (depth-field cueing) coded whether the translating field's depth-plane identity matched that of the delayed-onset field—that is, whether depth-plane membership tracked the temporal cue (1 = depth-cued, 0 = depth-uncued). F3 (color-field cueing) coded the analogous property for surface color (1 = color-cued, 0 = color-uncued). These three factors are fully crossed in the design—their combinations define the four swap conditions—and no prior observation was required to motivate their inclusion.

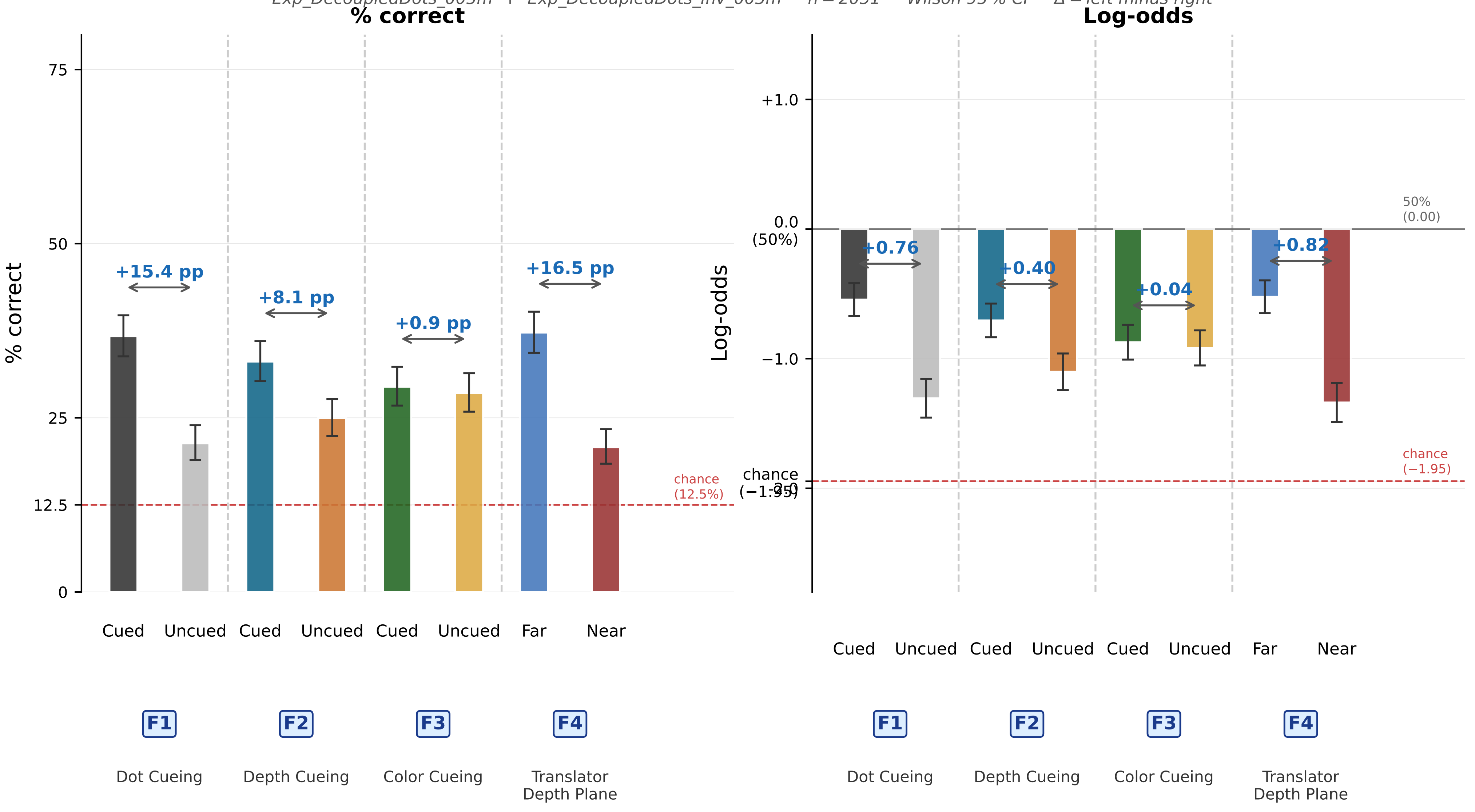
F4 (translator depth plane; 1 = Far, 0 = Near) was included as an empirically motivated covariate. Prior to this experiment, a robust Far > Near performance advantage was observed across independent sessions—parametric depth sweeps across a fourfold separation range and binocular-vs.-monocular comparisons—establishing a stereoscopic origin. Including F4 adjusts for this known source of variance and sharpens the cueing-factor estimates. The F4 asymmetry and its interactions are discussed on p. 7.

The model included all four main effects and three interaction terms motivated by the experimental logic. F1 × F2 tests the central question of the experiment: whether dot-onset cueing is effective only when the depth-plane identity of the translating field matches the identity established during the cue interval—that is, whether temporal and depth-plane cueing operate conjunctively rather than independently. F1 × F4 asks whether the dot-onset cueing benefit is itself depth-plane specific, operating more effectively at one stereoscopic depth than the other. F2 × F4 asks whether depth-field cueing is depth-plane specific—whether the attentional advantage conferred by matching depth-plane identity differs between the near and far planes. Color (F3) was entered as an additive term only; no interactions involving F3 were pre-specified, and none are reported. All predictors are binary (0 or 1). Coefficients were estimated by maximum likelihood, and effect sizes are reported as average marginal effects (AMEs) in percentage points, computed by averaging the predicted change in response probability across the observed distribution of covariates.

Coefficient estimates, AMEs, and conditional AMEs: see pp. 3–4.

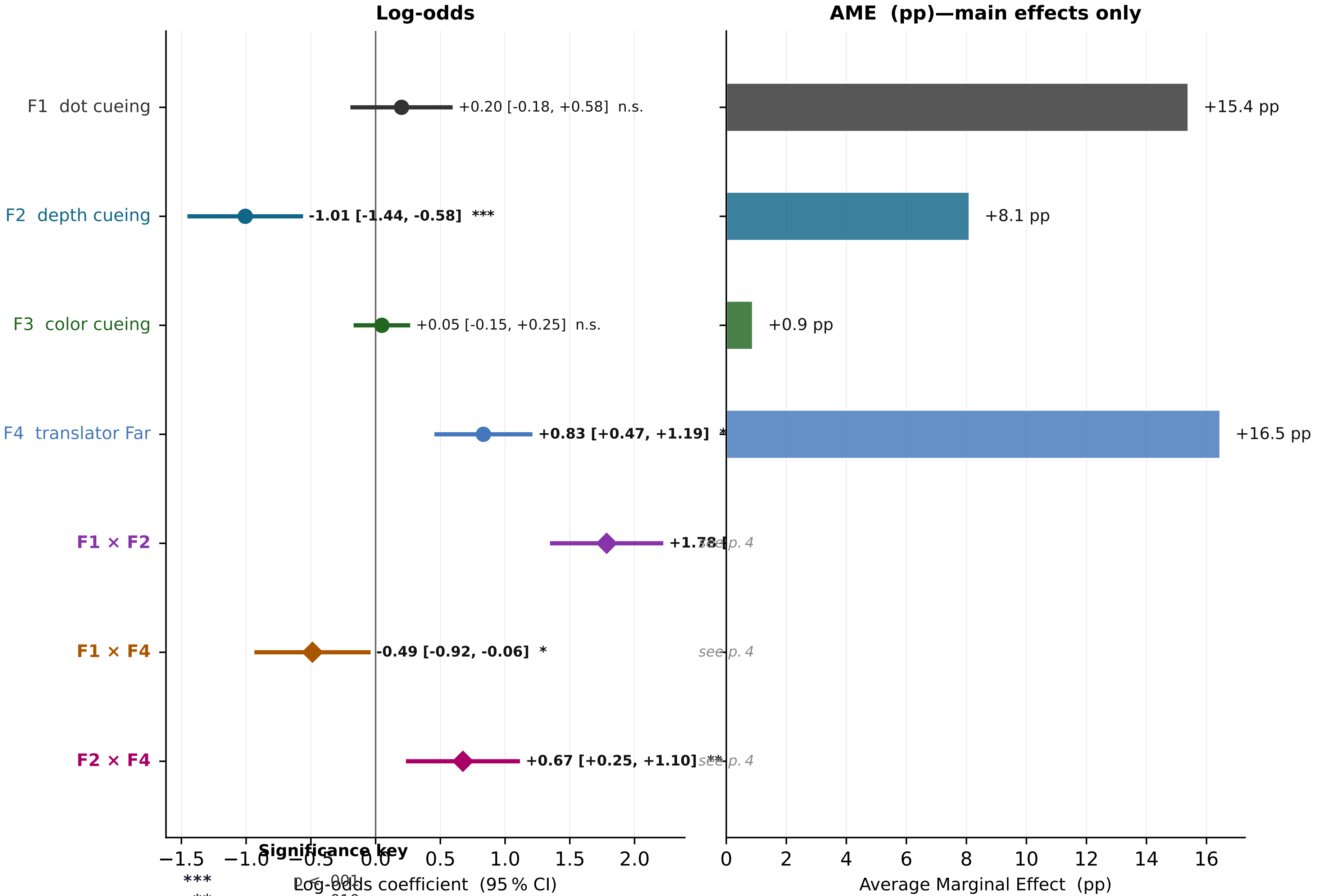
GLM predictors F1–F4 — performance by factor level

Exp_DecoupledDots_005m + Exp_DecoupledDots_Inv_005m · n = 2051 · Wilson 95 % CI · Δ = left minus right



Estimated GLM coefficients

Exp_DecoupledDots_005m + Exp_DecoupledDots_Inv_005m · n = 2051 · logistic regression · maximum likelihood



Significance key

***	p < .001
**	p < .010
*	p < .050
†	p < .100
n.s.	p ≥ .100

β0 (intercept) = -1.546 [-1.859, -1.232] p = 0.0000

≈ 17.6% predicted correct when all predictors = 0 (uncued, depth-uncued, color-uncued, Near translator)

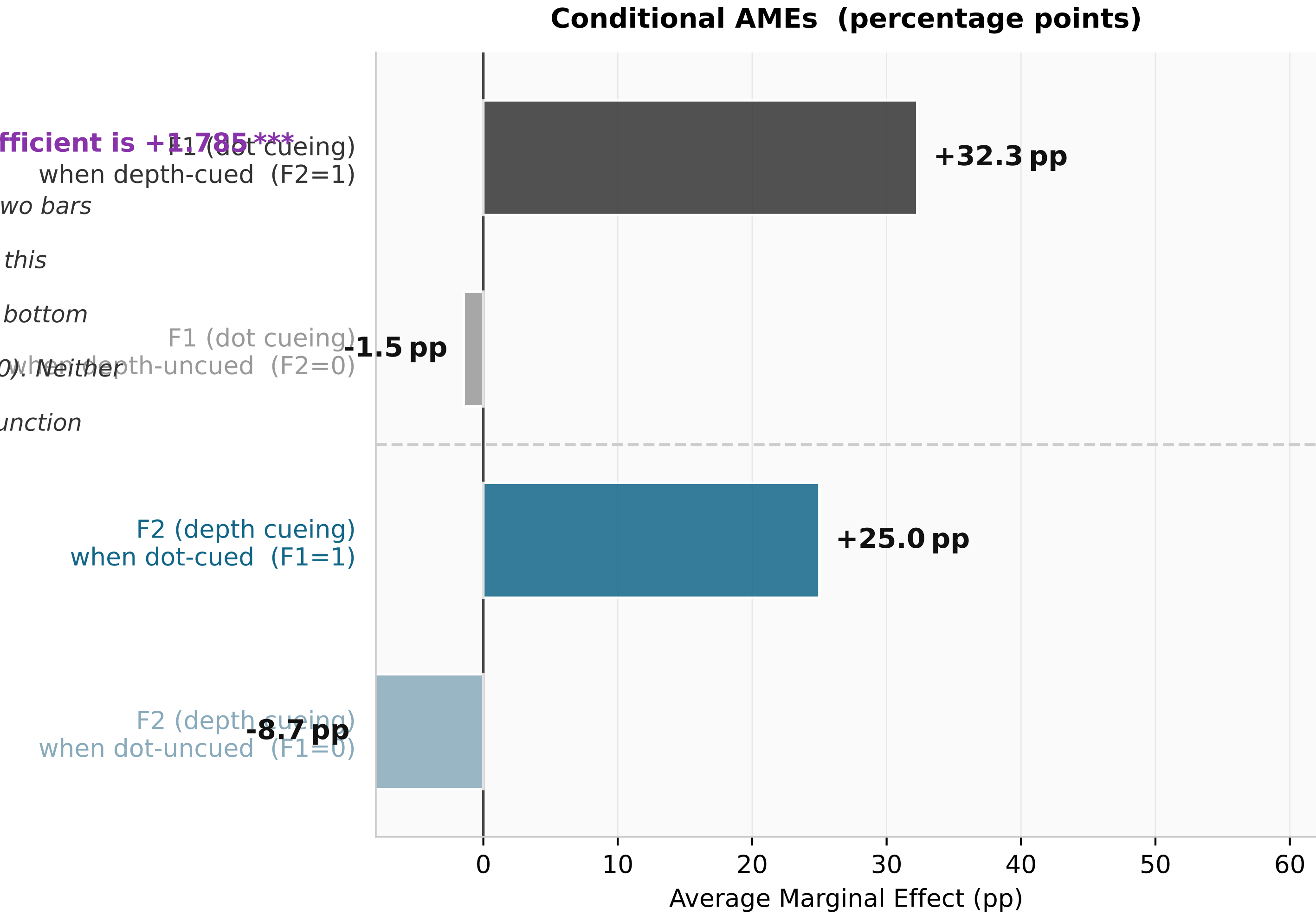
Conditional AMEs: why do F1 and F2 appear non-significant as main effects?

Exp_DecoupledDots_005m + Exp_DecoupledDots_Inv_005m · n = 2051

In an interaction model, β1 and β2 estimate conditional effects. β1 is the effect of F1 (dot cueing) specifically when F2 = 0 (depth-uncued trials); β2 is the effect of F2 specifically when F1 = 0 (dot-uncued trials). Because the temporal onset cue provides little benefit without depth-plane continuity, and depth-plane continuity provides little benefit without the onset cue, both of these conditional effects are small and may not reach significance individually. The full F1 × F2 interaction (β12 = +1.785, ***) captures the extra benefit when both factors are present together. The chart below shows the AME of F1 and F2 computed separately for each level of the complementary factor. The difference between the two F1 bars (or the two F2 bars) equals the interaction coefficient.

The F1×F2 interaction coefficient is +1.785 ***

The difference between the top two bars (F1 when F2=1 vs. F2=0) equals this coefficient directly. Same for the bottom two bars (F2 when F1=1 vs. F1=0). Neither cue alone is sufficient; their conjunction is.



Fitted equation and interpretation

Exp_DecoupledDots_005m + Exp_DecoupledDots_Inv_005m · n = 2051

$$\text{log-odds(correct)} = -1.546 + 0.200\cdot F1 + -1.007\cdot F2 + 0.048\cdot F3 + 0.833\cdot F4$$
$$+ 1.785\cdot(F1\times F2) + -0.487\cdot(F1\times F4) + 0.674\cdot(F2\times F4)$$

Predicted % correct for key conditions (F3 = 0 throughout)

Condition	F1	F2	F4	Log-odds	% correct
Uncued · depth-uncued · Near	0	0	0	-1.55	17.6%
Uncued · depth-uncued · Far	0	0	1	-0.71	32.9%
Cued · depth-uncued · Near	1	0	0	-1.35	20.7%
Cued · depth-uncued · Far	1	0	1	-1.00	26.9%
Cued · depth-cued · Near	1	1	0	-0.57	36.2%
Cued · depth-cued · Far ★ optimal	**1**	**1**	**1**	**+0.45**	**61.1%**

The fitted equation makes the conjunctive structure of the result explicit. When the temporal onset cue is present but the translating field's depth-plane identity has been disrupted ($F1 = 1$, $F2 = 0$), the model predicts modest performance only marginally above the uncued baseline. When both the temporal cue and depth-plane continuity are intact ($F1 = 1$, $F2 = 1$), predicted performance is approximately double that level. The $F1 \times F2$ coefficient is large and positive, capturing a synergistic effect: the two cues are not simply additive but jointly necessary for robust surface selection. Neither alone is sufficient; both together are.

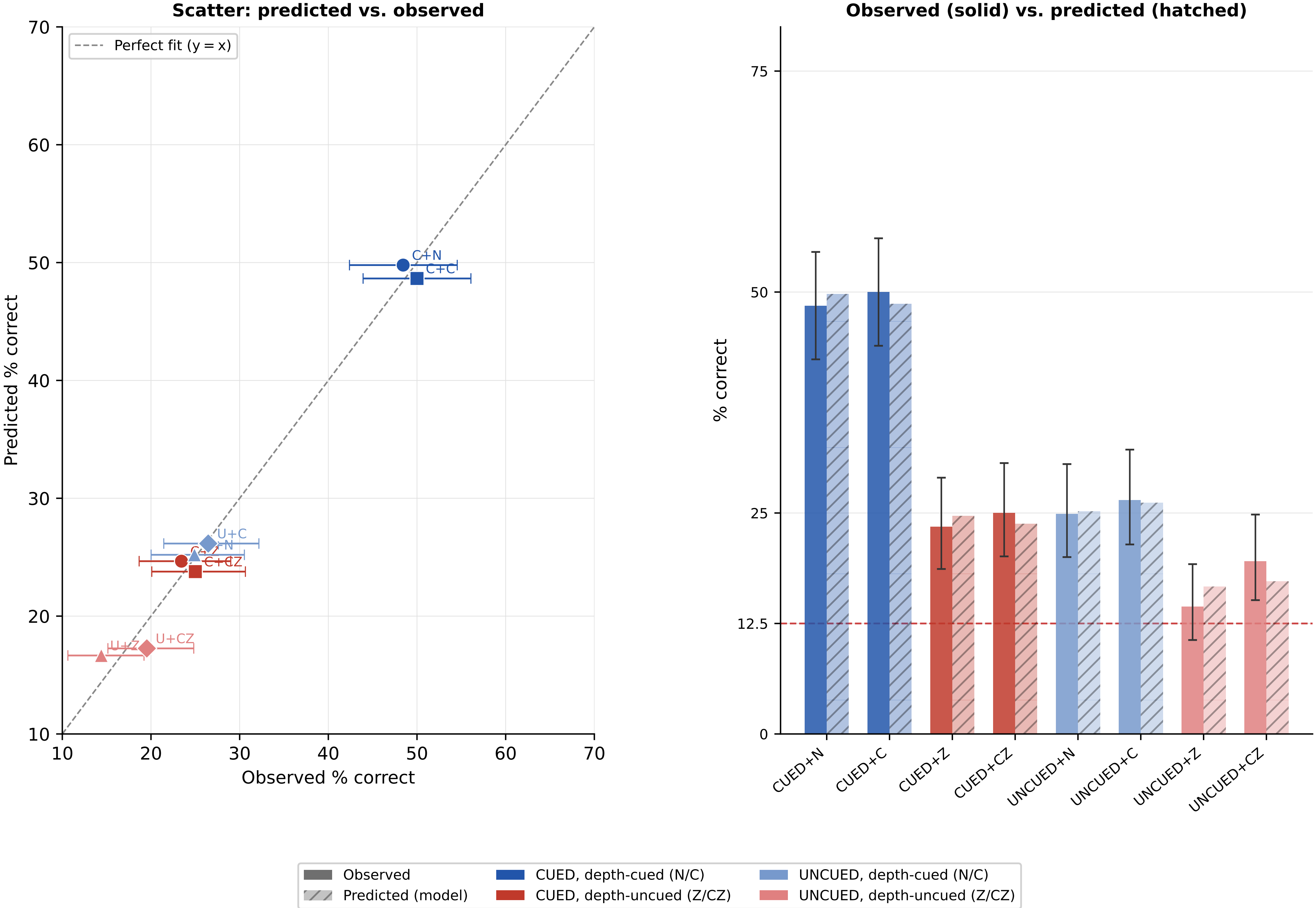
The null result for color ($F3 \approx 0$) resolves an ambiguity in earlier experiments using linked depth and color, where Near was always Red and Far always Green. In those sessions an apparent color-cueing effect was observed, but the present design—in which color and depth swap independently—shows that effect to be entirely a depth confound. Color-field identity does not bind to the temporal onset cue.

The predicted performance when $F1 = 1$ and $F2 = 0$ (temporal cue present, depth disrupted) provides a natural point of contact with earlier two-dimensional transparent-surface experiments. In those studies, overlapping random-dot fields sharing a single depth plane were used; no stereoscopic depth information was available, and the temporal onset cue operated without any depth-plane binding. The performance level we observe here under depth disruption is consistent with that two-dimensional baseline, suggesting that the temporal onset mechanism is preserved in VR and that the improvement seen with depth continuity reflects an additional contribution of stereoscopic object identity rather than a change in the basic cueing mechanism. Depth-plane membership appears to function as a constitutive feature of the attentional object representation: its disruption at the moment of translation partially disassembles the representation that the temporal cue had established.

F4 (translator depth plane; 1 = Far, 0 = Near) shows a robust Far > Near advantage of approximately +15 pp ($p < .001$). Whether this reflects a Far-plane advantage or a Near-plane penalty cannot be determined from Near/Far comparisons alone: a flat baseline (zero disparity) would be needed to locate the asymmetry. The monocular control sessions are the closest available reference: the Far > Near gap is absent monocularly, suggesting binocular disparity selectively benefits the Far condition. The asymmetry also grows with stereoscopic separation rather than shrinking, which argues against a simple near-plane leakage account (more discriminable planes should reduce interference, not increase it). $F1 \times F4$ and $F2 \times F4$ interactions are shown on p. 7. Mechanism unresolved.

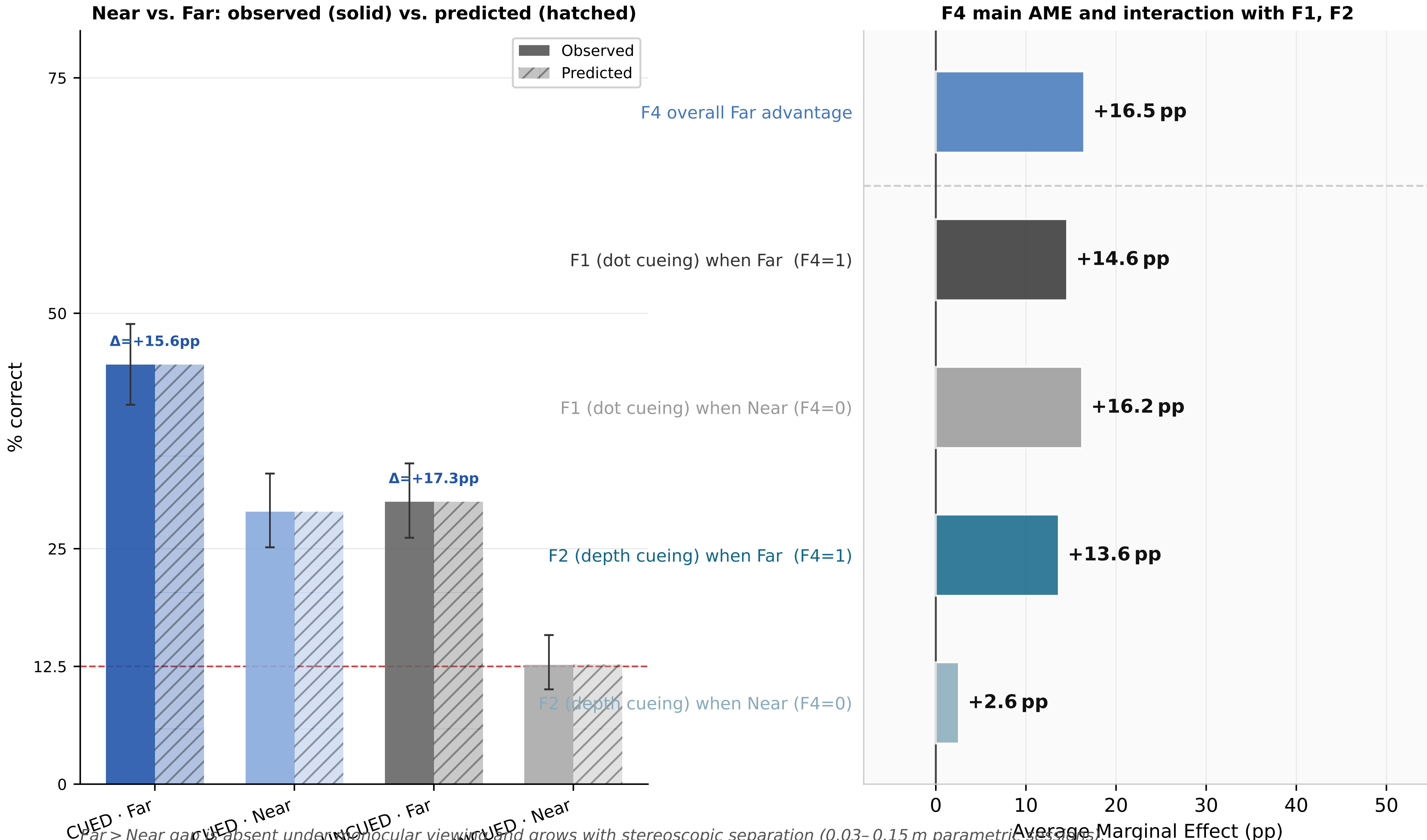
Model fit: predicted vs. observed accuracy

Exp_DecoupledDots_005m + Exp_DecoupledDots_Inv_005m · n = 2051 · 8 conditions (swap × cueing)



F4 — Far > Near asymmetry: performance, interactions, and open questions

Exp_DecoupledDots_005m + Exp_DecoupledDots_Inv_005m · n = 2051



Far > Near gap is absent under monocular viewing and grows with stereoscopic separation (0.03– 0.15 m parametric sessions).

The direction of the effect (Far advantage vs. Near penalty) cannot be resolved without a zero-disparity baseline.

Leakage account predicts the gap should shrink as planes become more discriminable; it grows instead. SOA manipulation is the planned critical test.